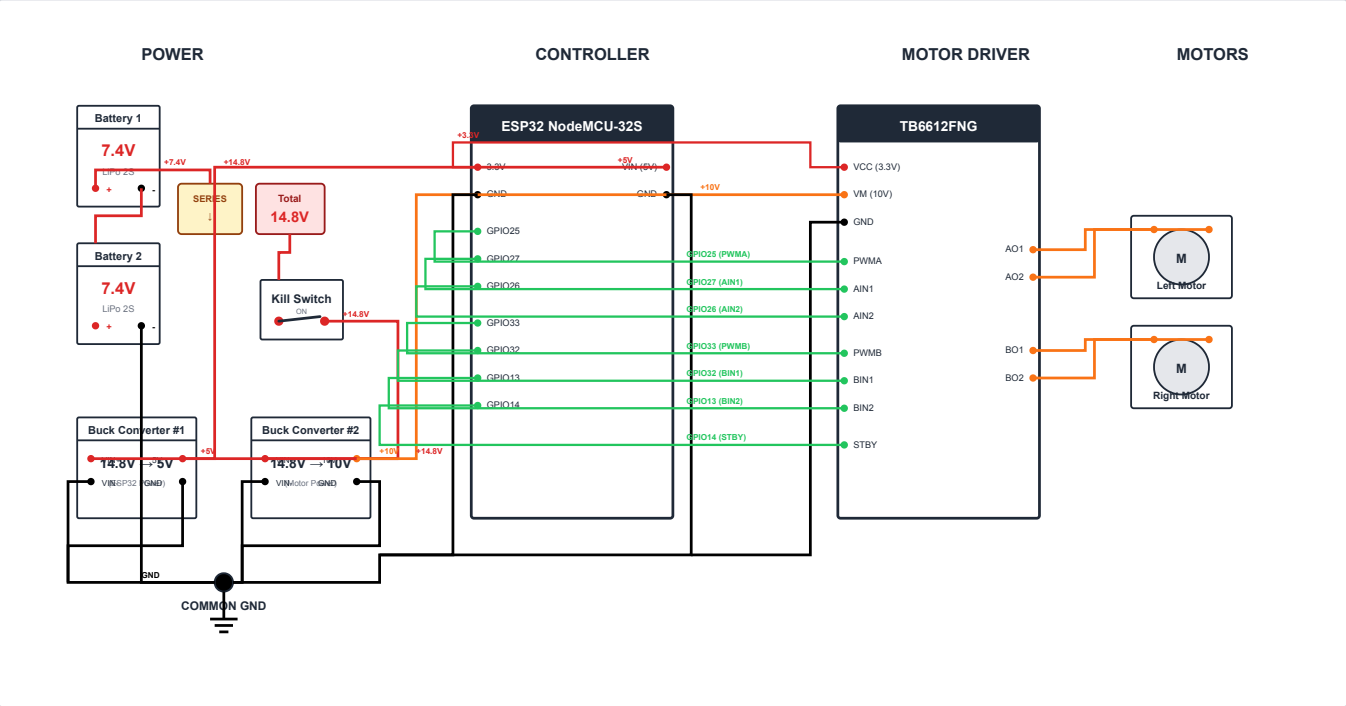
ESP32 Sumo Bot Wiring Diagram

Series battery configuration with dual buck converters



Battery Power (+)

Motor Power

Ground (-)

GPIO/Control

Power Architecture

- 2x 7.4V LiPo batteries in SERIES
- Total voltage: 14.8V
- Kill switch on main power line
- Buck #1: 14.8V → 5V (ESP32)
- Buck #2: 14.8V → 10V (Motors)
- ESP32 3.3V logic from internal regulator
- Common ground required for all components

Motor Driver (TB6612FNG)

- VCC: 3.3V from ESP32
- VM: 10V from Buck Converter #2
- PWMA: GPIO25 (left motor speed)
- AIN1/AIN2: GPIO27/GPIO26 (left direction)
- PWMB: GPIO33 (right motor speed)
- BIN1/BIN2: GPIO32/GPIO13 (right direction)
- STBY: GPIO14 (standby control)
- AO1/AO2 → Left Motor
- BO1/BO2 → Right Motor

Component List & Safety

Components:

- ESP32 NodeMCU-32S
- TB6612FNG motor driver
- 2x DC motors (6-12V)
- 2x 7.4V LiPo batteries
- 2x Buck converters
- Kill switch

Safety:

- Verify series battery polarity
- Kill switch must be accessible